

Note on the Replicability of Miller and Weinberg (2017)

Nathan Miller and Matthew Weinberg

September 30, 2021

Earlier this month we discovered that the random sampling procedure used in Miller and Weinberg (2017) to generate consumer draws from the income distribution does not enable replication.¹ That is, rerunning the code provided does not replicate the consumer draws that are used for the article. The statistical distribution of any sample of draws created with the code should be the same, though as the samples are finite, the actual draws should vary. There is no reason aside from random sampling that the draws used in the article would be “unusual” any way—the issue is purely replicability.

This document discusses the role of the consumer draws in the analysis, analyzes whether the draws that we used in the study are statistically unusual, and shows why the random sampling procedure does not enable replication.

The Role of the Consumer Draws

The article documents that the prices of the Miller, Coors, and ABI increase after the MillerCoors joint venture is consummated in July, 2008. This is consistent with the joint venture having anticompetitive price effects, although of course there are other possible explanations, including the onset of the Great Recession. One thing that stands out about the pricing patterns is that the prices of ABI increase by about the same amount as the prices of Miller and Coors. This would not be predicted by standard models of differentiated-products Bertrand competition, but it is consistent with other forms of competition, such as price coordination.

To make progress, the article explores the possibility that the MillerCoors joint venture facilitated coordination, using a structural model of demand and supply that allows coordination to deepen once the joint venture is consummated. On the demand-side, the article uses a nested logit model and four different specifications of a random coefficients logit model; similar results obtain for each.

The consumer draws are used in the random coefficients logit models. In particular, consumers of different incomes are allowed to differ in how they view the trade-off between price and other product characteristics. For data on the consumers, a sample of households is obtained from the PUMS data of the American Community Survey, as described in the article. For each year of the sample (2005-2011) and each of 39 regions, a random sample of 500 households is drawn from the PUMS data. Then, with the draws in hand, estimation proceeds with the standard nested-fixed point approach.

The draws are helpful in allowing some products to be closer substitutes than others. However, identification on the supply-side of the model (and particularly inferences about price coordination) comes primarily from the price trends described earlier, rather than from the consumer draws. Furthermore, comparable results obtain with the nested logit specification, which does not use the consumer draws. Therefore, we think of the consumer draws as playing a limited role in the empirical analysis, and do not view them as critical for our main conclusions.

¹ Miller, Nathan and Matthew Weinberg (2017). “Understanding the Price Effects of the MillerCoors Joint Venture,” *Econometrica*, 85(6): 1763-1791.

A Monte Carlo Analysis of the Draws

We have conducted a Monte Carlo analysis to explore whether the draws that we use in the article are unusual. To implement, we created 100 alternative data sets, each containing 500 draws of income for each year and region in our sample. We then computed the mean and standard deviation of income for the draws in each data set, year, and region. Finally, we compare these to the mean and standard deviation of income of the draws used in the article.²

We find that, across the 273 region*year combinations, the 90th percentile range of means from the 100 alternative data sets contains the mean of the actual draws used in our paper 89 percent of the time. Further, the 90th percentile range of standard deviations from the 100 alternative data sets contains the mean of the actual draws used in our paper 88 percent of the time. With randomization, those percentages should converge to 90 percent as the number of alternative data sets grows large, so we interpret the results as supporting that the actual draws are in line with what one should expect from a random sample procedure. Based on this finding, and our interpretation of the role that the draws play in our analysis, we conclude that the results in the article are robust despite the replicability problem.

Later in this document we provide two figures, for the mean and standard deviation, respectively, that show that the actual draws fall within the ranges that we obtain with the 100 alternative data sets. The figures break out each of the regions for the year 2008; results for other years are similar.

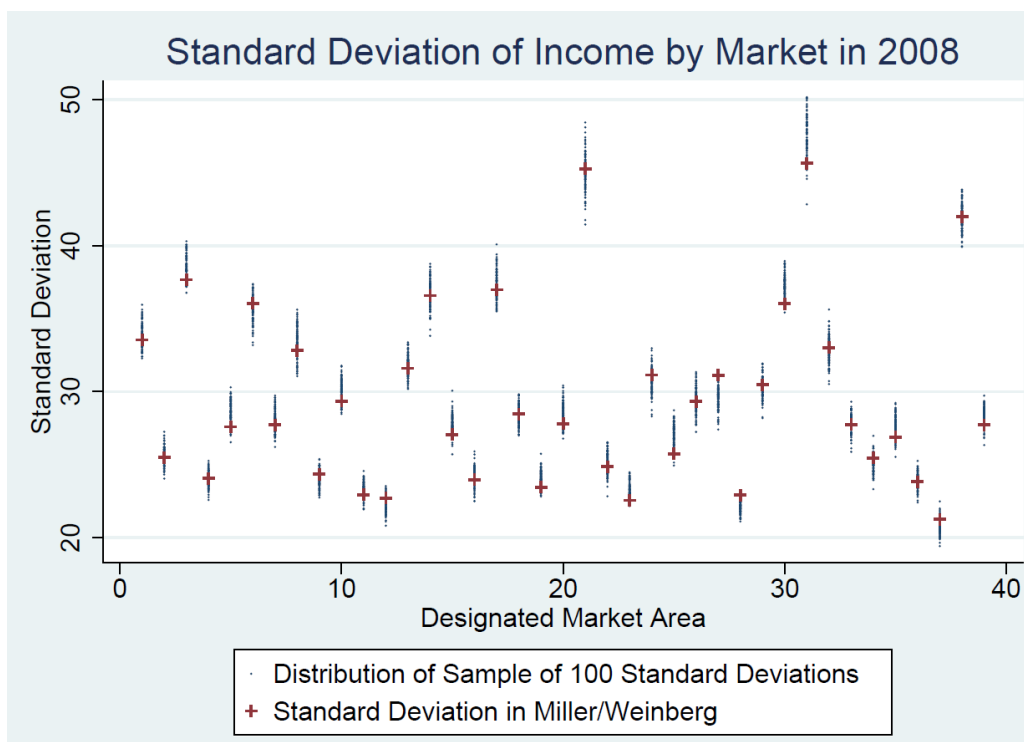
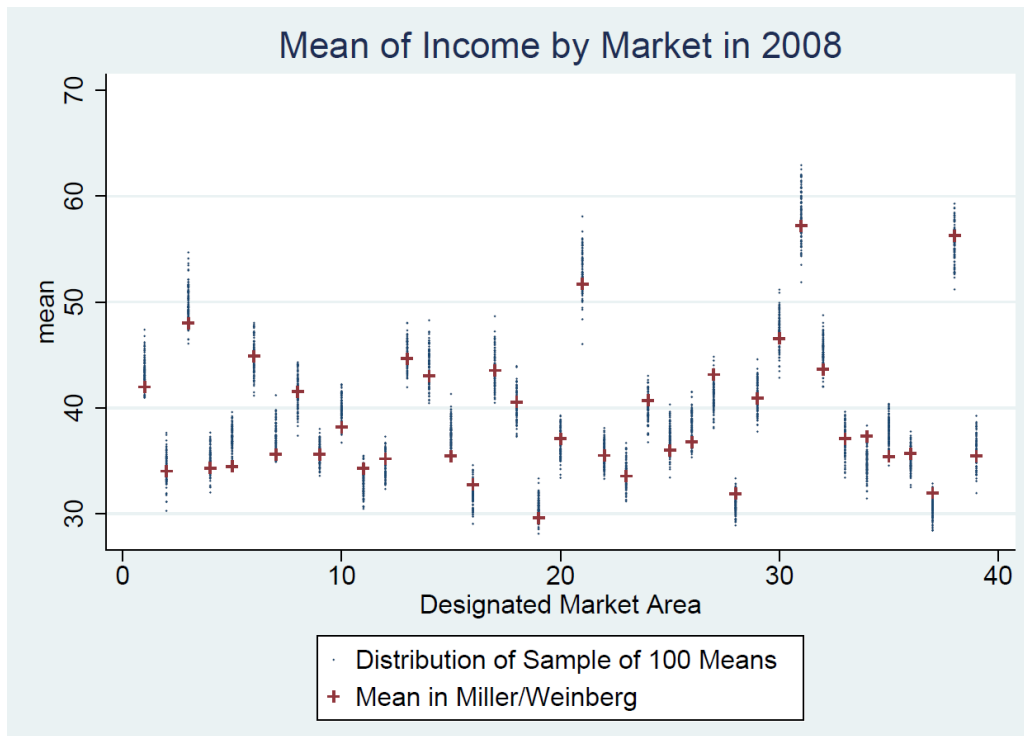
Stata Code and Replicability

We now describe the reason that code does not allow for replication of the consumer draws. For background, the Stata code at issue is contained in the file “PUMS Demo Creation.do” that is available, along with the rest of our code, on the *Econometrica* website. The structure of the Stata code can be summarized as follows:

- Set the seed (we use “`set seed 123456789`”).
- Read in the PUMS data.
- Conduct an m:1 merge that brings in an alternative region identifier.
- Sample from the PUMS data using the `gsample` command.
- Save the results as `demosE.csv`.

What we did not recognize when writing the code is that Stata sorts observations randomly after any m:1 merge. Thus, even though the m:1 merge is well-defined, and everything matches as it should, the ordering of observations that comes out of the merge is random. Furthermore, the randomization is not controlled by the “`set seed`” command. To replicate the sorting, we would have needed to use the “`sortseed`” command. Because we neglected to do so, the order of observations is randomized before sampling occurs. Thus, even though the sampling itself controlled by the “`set seed`” command, the draws that come out of the program do not replicate. We have included a figure below that shows some of the code in the Stata file, with annotations that clarify how the code operates.

² We have designed the Monte Carlo analysis so that it is replicable. The authors can provide code upon request.



```

179 * "E" = non-negative HH income normalized by HH size, top-coded at 95th percentile
180 use "$PUMS/PUMS Cleaner H Output", replace
181 sort iri
182 merge m:1 iri using "$PUMS/mktmatch" }
183 drop _merge iri
184 rename (hRealInc span age60 wgt) (realInc hisp age pwgtp) }
185 replace realInc = realInc/np
186 by year market, sort: egen topInc = pctlc(realInc), p(95)
187 replace realInc = topInc if realInc > topInc
188 drop if realInc < 0
189 save "$PUMS/temp0", replace }
190 mktlooper1
191 use "$PUMS/draws", replace }
192 order year market realInc* age* hisp*
193 sort year market
194 save "$PUMS/demosE", replace
195 outsheet year market realInc* age* hisp* using "$PUMS/demosE.csv", comma names replace

```

The purpose of this merge is to bring in a different identifier for the region. However, after a m:1 merge, Stata randomly sorts the observations, and the randomization is not controlled by Stata's set seed command. Therefore, we lose replicability here.

Normalizing household income by household size and top-coding. This is fine.

The last lines sort the draws obtained with the sampling procedure and save the output as demosE.csv, which then is used in Matlab to estimate random coefficients logit models of demand.

Calls a function (mktlooper1) that conducts the sampling with `gsample`. The seed has been set earlier in the program so this step is okay for replication. However, due to the m:1 merge earlier, the order of the data from which draws are taken changes randomly with each attempt to replicate, in a way that is not controlled with Stata's set seed command.